

# **PROJECT TITLE:- EV CHARGING STATION PERFORMANCE ANALYSIS**

## **1. Project Overview**

The EV Charging Station Performance Analysis is a Power BI-based data analytics project designed to analyse **electric vehicle charging station performance** and identify important patterns related to station utilization, charging demand, port availability, pricing, charging power, waiting time .

The dashboard provides an interactive view of EV charging station data and helps users understand station performance, charging demand patterns, pricing behaviour and operational conditions across different stations, charger types, locations and time periods.

## **2. Objective**

The main objective of this project is to analyse EV charging station data and identify important operational and demand-related patterns.

The dashboard helps users:

- Analyse charging station utilization.
- Understand charging demand across different charger types.
- Compare available, occupied and out-of-service ports.
- Analyse charging behaviour across different hours.
- Identify areas where station performance can be improved.

## **Tools Used**

- **Microsoft Power BI**
- **Power Query**
- **DAX**
- **Data Cleaning & Transformation**
- **Data Analysis & Visualization**
- **Interactive Slicers & Filters**

## **3. Key Insights**

### **1. Charging Sessions**

The dashboard provides an overview of the total charging sessions recorded in the dataset and helps understand the overall charging demand.

### **2. Station Utilization**

Station-wise utilization analysis helps identify stations experiencing higher charging activity and differences in station performance.

### 3. Charger Type

Charging sessions vary across different charger types, helping understand the demand for different charging technologies.

### 4. Port Availability

The dashboard compares available, occupied and out-of-service ports to understand charging infrastructure availability.

### 5. Time & Demand

Hourly utilization and peak vs non-peak analysis help identify changes in charging demand throughout the day.

### 6. Session Duration

Average session duration provides an understanding of how long charging sessions typically remain active.

### 7. Pricing

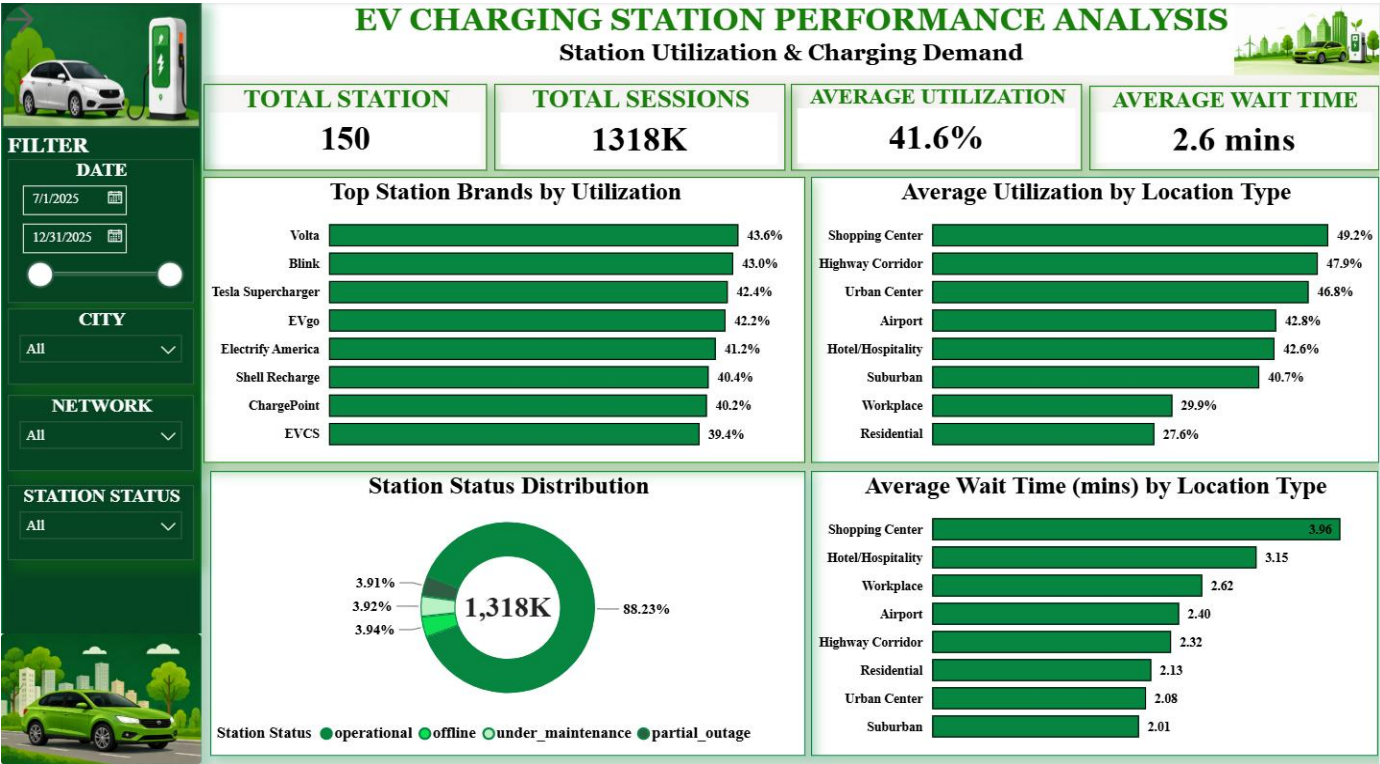
Pricing analysis compares charging activity and pricing patterns across different pricing types and charging networks.

### 8. Traffic Conditions

The average traffic congestion index provides an overview of traffic conditions around the charging stations.

## 4. Analysis

### 4.1 Executive Overview

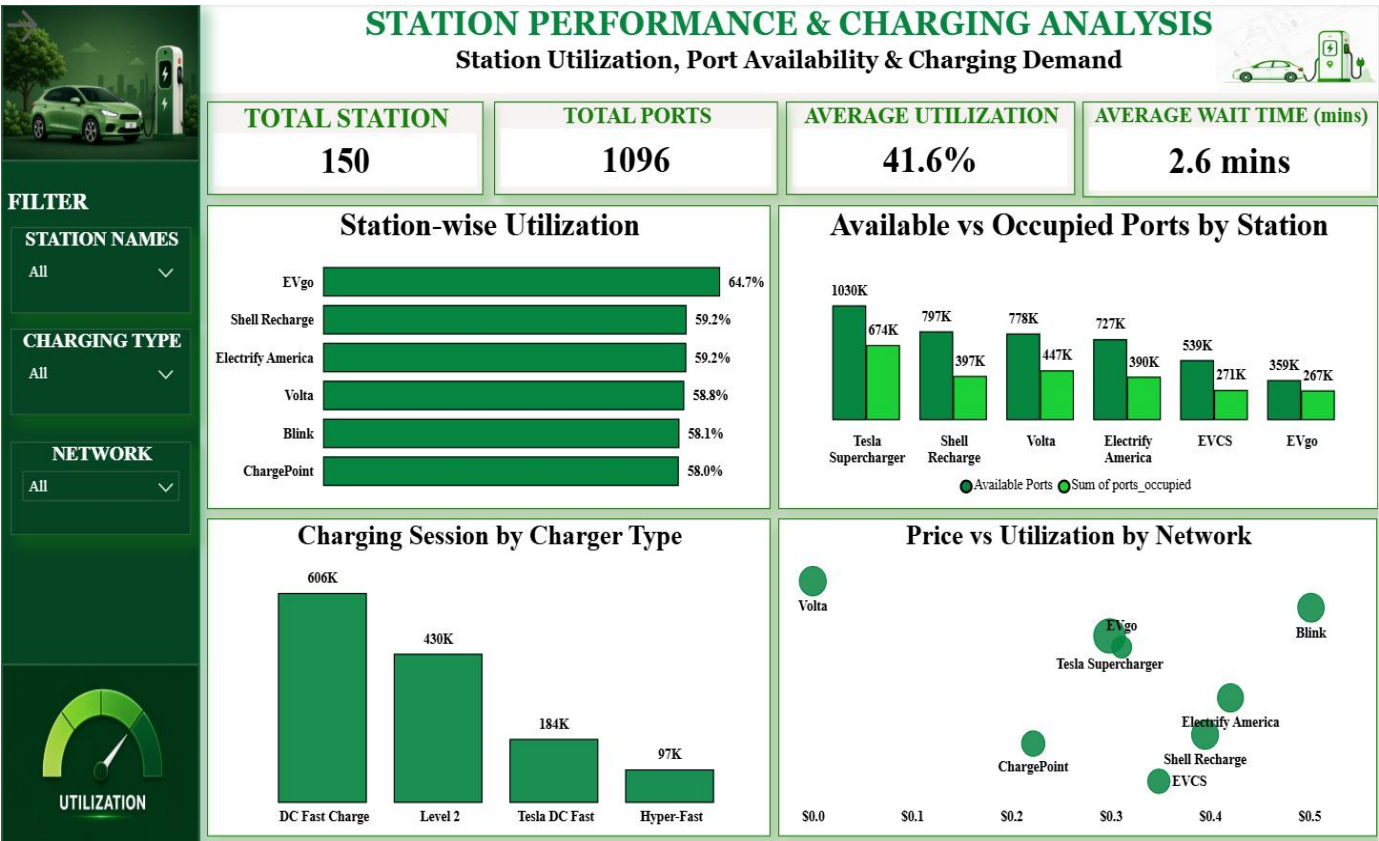


The Executive Overview provides a high-level view of EV charging station performance. It presents important KPIs and visualizations related to station utilization, charging sessions, location types, station status and waiting time.

Key Analysis

- Provides an overall view of station performance.
- Compares utilization across location types.
- Highlights station operational status.
- Helps identify stations with different utilization and waiting-time patterns.

4.2 Station Performance & Charging Analysis

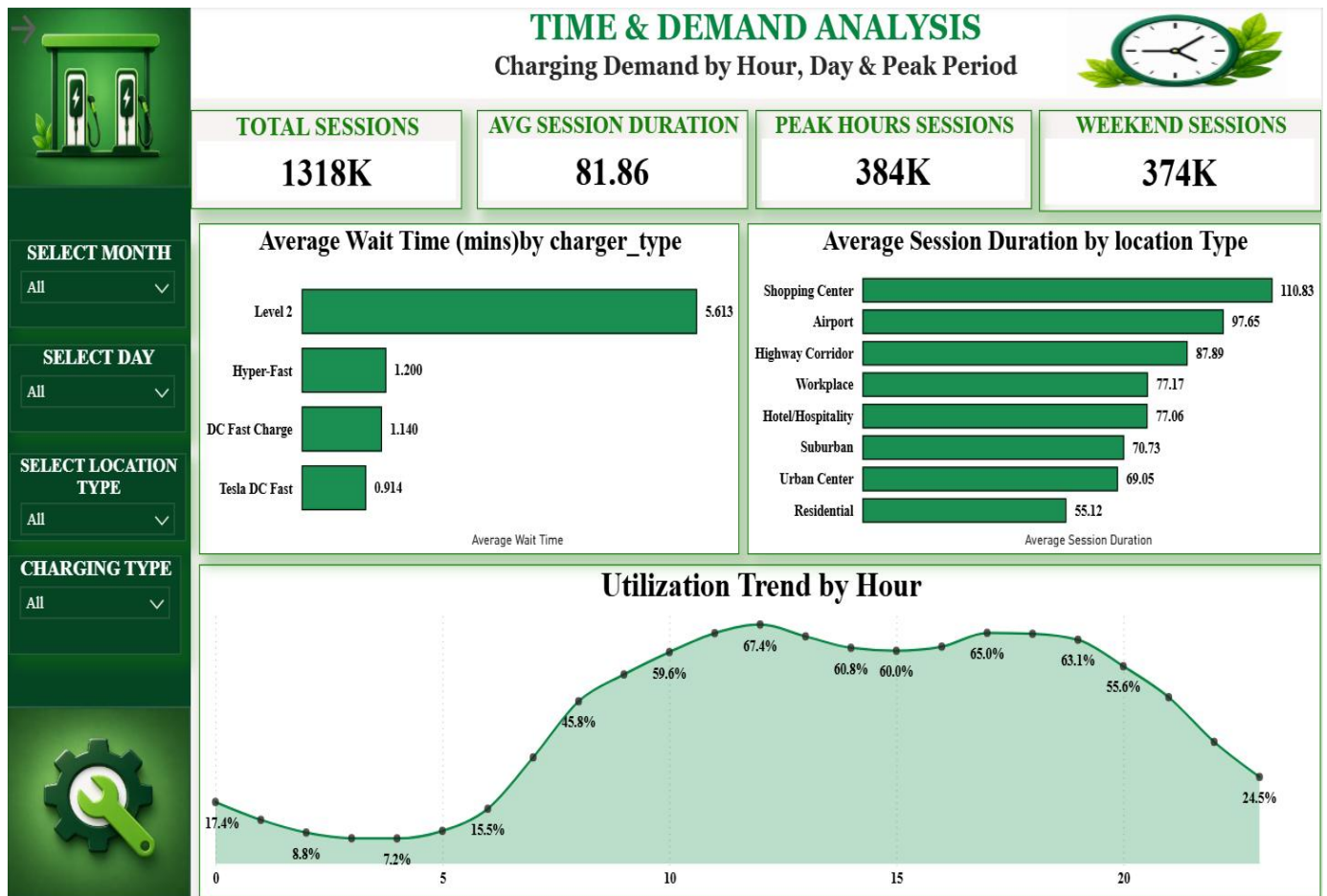


This page focuses on station-level performance and charging infrastructure.

Key Analysis

- Compares utilization across stations.
- Shows available ports by station.
- Compares available and occupied ports.
- Shows charging activity across different charger types.
- Helps to understand Price Vs Utilization by Network.

## 4.3 Time & Demand Analysis



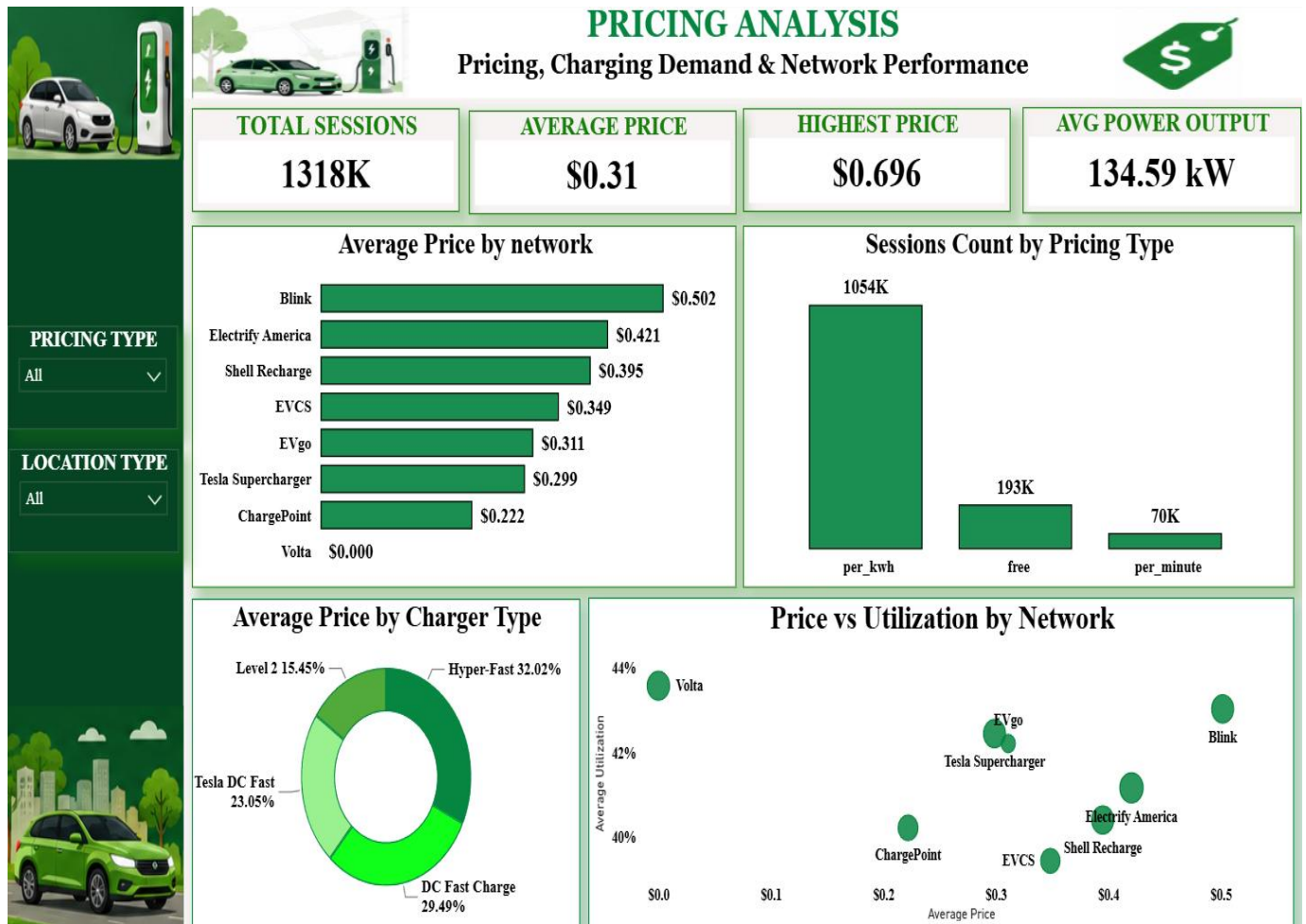
This page analyses charging demand and utilization across different time periods.

### Key Analysis

- Shows total charging sessions.
- Analyses average session duration.
- Compares peak and non-peak charging sessions.
- Analyses utilization by hour.
- Compares waiting time across charger types.
- Shows session duration across different location types.

This analysis helps understand when charging activity is higher and how charging behaviour varies throughout the day.

## 4.4 Pricing Analysis



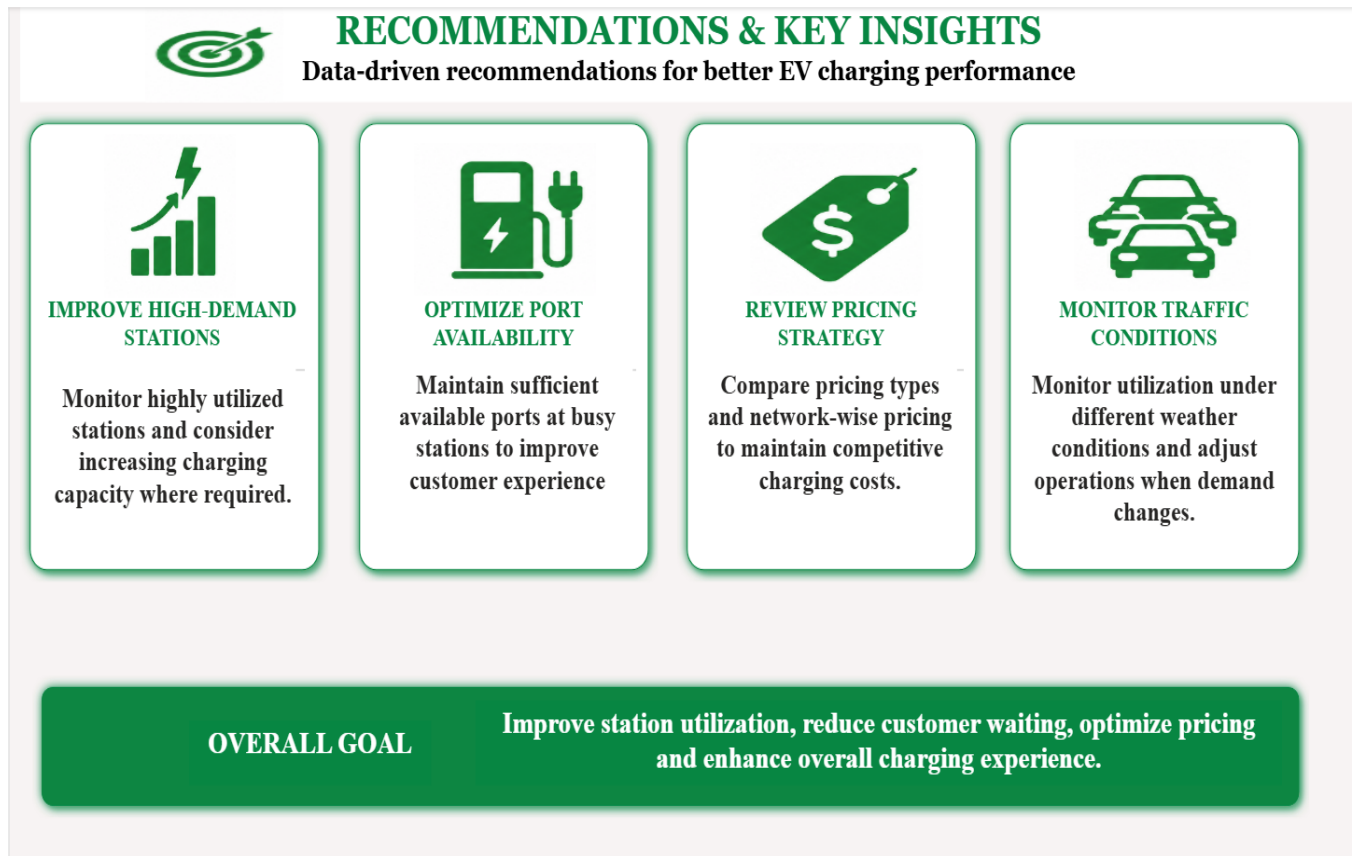
The Pricing Analysis page focuses on charging prices and related charging patterns.

### Key Analysis

- Analyses charging sessions by pricing type.
- Compares pricing patterns across the available categories.
- Shows average charging price.
- Analyses average power output.
- Provides an overview of traffic congestion around charging stations.

This helps understand the relationship between charging activity, pricing and operating conditions.

## 4.5 Recommendations & Key Insights



The final page summarizes the major findings and provides practical recommendations based on the dashboard analysis.

### Recommendations

#### 1. Improve High-Demand Stations

Monitor highly utilized stations and consider increasing charging capacity where required.

#### 2. Optimize Port Availability

Maintain sufficient available ports at busy stations to improve customer experience.

#### 3. Review Pricing Strategy

Compare pricing types and network-wise pricing to maintain competitive charging costs.

#### 4. Monitor Traffic Conditions

Use traffic patterns to understand demand and improve station planning during busy periods.

### Overall Goal

Improve station utilization, reduce customer waiting, optimize pricing and enhance the overall EV charging experience.

## 5. Conclusion

The **EV Charging Station Performance Analysis Dashboard** provides an interactive and visual representation of EV charging station data. It helps identify important patterns related to station utilization, charging demand, port availability, session duration, pricing, power output and traffic conditions.

The analysis provides a better understanding of how charging stations perform across different stations, charger types, locations and time periods. It also highlights operational areas such as port availability, charging demand and pricing that can be monitored for better station management.

Overall, this project demonstrates how **Microsoft Power BI, Power Query and DAX** can be used to transform raw EV charging data into meaningful insights and support **data-driven decision-making for EV charging infrastructure management**.